

NLP X CSS · SEMINAR PROJECT

Assertion and Question Developer Agents

A prompt-driven pipeline that turns raw indicators into theory-grounded survey questions, evaluated against an expert gold set.

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AGENDA

1 · The Problem

Why survey-item writing is hard expert work

2 · The Framework & Data

Annotation theory + 115-row gold set

3 · Methodology

The pipeline, the model, and **why** each choice

4 · Key Findings

Four runs, figures, where it wins & fails

5 · Discussion

What it means, limitations

6 · Conclusion

Takeaways + next steps + Q&A

CONTEXT

Our agents inside the full pipeline



Our contribution

We design, implement, and evaluate the **Assertion Developer** and **Question Developer** agents: the core measurement-decision layer of the pipeline.

Evaluation strategy

Each agent is scored **independently**: the Question Developer always receives the **gold assertion**, so errors don't compound across stages.

Writing good survey questions is hard, manual, expert work

- Social science runs on surveys, but a **bad question gives invalid data**.
- Researchers start from a vague **indicator** (e.g. "satisfaction with care") and must turn it into a precise, neutral, theory-grounded question.
- This requires applying a **strict measurement framework** consistently across hundreds of items: slow and expensive.

Our research question

Can an **open LLM**, guided only by prompts, reproduce this expert process: picking the right measurement category **and** writing the question?

If yes, it accelerates questionnaire design and makes it more consistent and reproducible.

A 3-step recipe: Indicator → Assertion → Question

① Indicator

Raw idea to measure



② Assertion

Precise statement
+ **concept** + **structure** tag



③ Question

Respondent wording
+ answer options

22 Basic Concepts

What is being measured: e.g. Evaluation, Behaviour, Preference, Demographics. Split into **subjective** (14) vs **objective** (8).

3 Semantic Structures

The grammar of the assertion, written as short **codes**: xIe , rDy , xPRy ... Each concept allows only certain codes.

Think of it as **grammatical parsing for survey items**: first classify the item, then build it to a template.

MAKE IT CONCRETE

One indicator, all the way through

"Overall satisfaction
with product"

indicator



Concept: Evaluation
Structure: xle

"I am satisfied with this product."



"How satisfied are you
with this product?"

Rating scale · Direct interrogative

Why the middle step matters

The assertion forces the model to **commit to a measurement decision** before writing wording. That decision is where difficulty, and most errors, live.

A near-miss: Repurchase intention

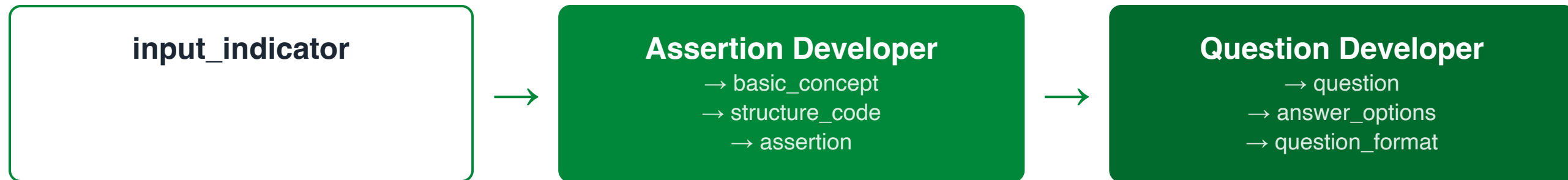
Gold: Action tendencies · rFDy

Model: Action tendencies · rFDy ✓ → **"How likely are you to purchase again?"**

Both codes express future behavior: the model must distinguish **intention** (rFDy) from **forecast** (xFDy).

TASK DEFINITION

Two agents, two sub-tasks



ASSERTION DEVELOPER PROMPT RULES

- Pick exactly **one** of 22 concepts (enum-constrained)
- Choose only structure codes **allowed** for that concept
- Apply disambiguation table for confusable pairs
- Output strict JSON: `concept · structure_code · assertion`

QUESTION DEVELOPER PROMPT RULES

- Convert assertion `→` direct interrogative question
- Subjective concepts `→` rating scale; objective `→` frequency / open-ended
- Options must be exhaustive and mutually exclusive
- Output strict JSON: `question · answer_options · question_format`

Both run on `Qwen3.5-9B` with `enable_thinking=false`; the concept rulebook is injected from `concepts.yaml` so prompt and rules stay in sync.

The gold set: 115 expert-annotated rows

- **115 rows**, covering **all 22 concepts**.
- Each row: indicator, gold concept, gold structure code, gold assertion, gold question + options, domain.
- This is our **ground truth** for scoring.

Heavily imbalanced

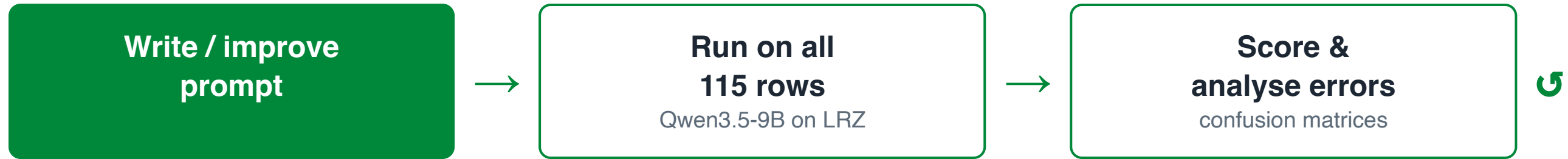
Evaluation (20), Demographics (15), Preference (10) dominate. **Many concepts have only n = 3.**

Concept	n
Evaluation	20
Demographics	15
Preference	10
Behaviour / Cognitive judg. / ...	7
Action tendencies / Events	5
Feelings	4
13 other concepts	3 each

Key consequence: 115 rows is enough to **evaluate** a model, but too few to reliably **fine-tune** one → motivates a prompting-first approach.

METHODOLOGY

A tight prompt-engineering loop



Three **prompt-engineering iterations** (Runs 1–3), each targeting the biggest error category from the last run, with **no model training**. **Run 4** re-scores Run 3's exact outputs with an **LLM-as-judge** to measure semantic quality.

1

INITIAL ZERO-SHOT

2

PROMPT TUNING

3

STRUCTURE FOCUS

4

+ SEMANTIC JUDGE

Why these tools?

Choice	Why we made it
Prompting before fine-tuning	Only 115 examples, too few to train reliably . Prompting is cheap, fast to iterate, and interpretable : we can read exactly why a choice was made. It sets an honest baseline before any training.
Structured JSON output (guided decoding + concept <code>enum</code>)	Guarantees machine-parseable output and forces the model to pick a valid concept from the 22: it cannot invent categories.
<code>enable_thinking = false</code>	Qwen3.5 otherwise emits long reasoning prose that breaks JSON parsing . Disabling it gave reliable structured answers.
Isolated evaluation	Score each agent independently : the Question Developer is fed the gold assertion , not the model's own. Errors don't compound, so we can localise failures.
Normalization layer	Fair scoring: unify <code>Behavior=Behaviour</code> , extract the structure code from messy text, without ever editing the gold file .

Evaluation metrics

Assertion stage

- **Concept accuracy**: right category?
- **Structure accuracy**: right code?
- **Both correct**: the strict bar

Question stage

- **Coverage**: produced a non-empty question?
- **Exact match** vs gold wording
- **Question format** classified

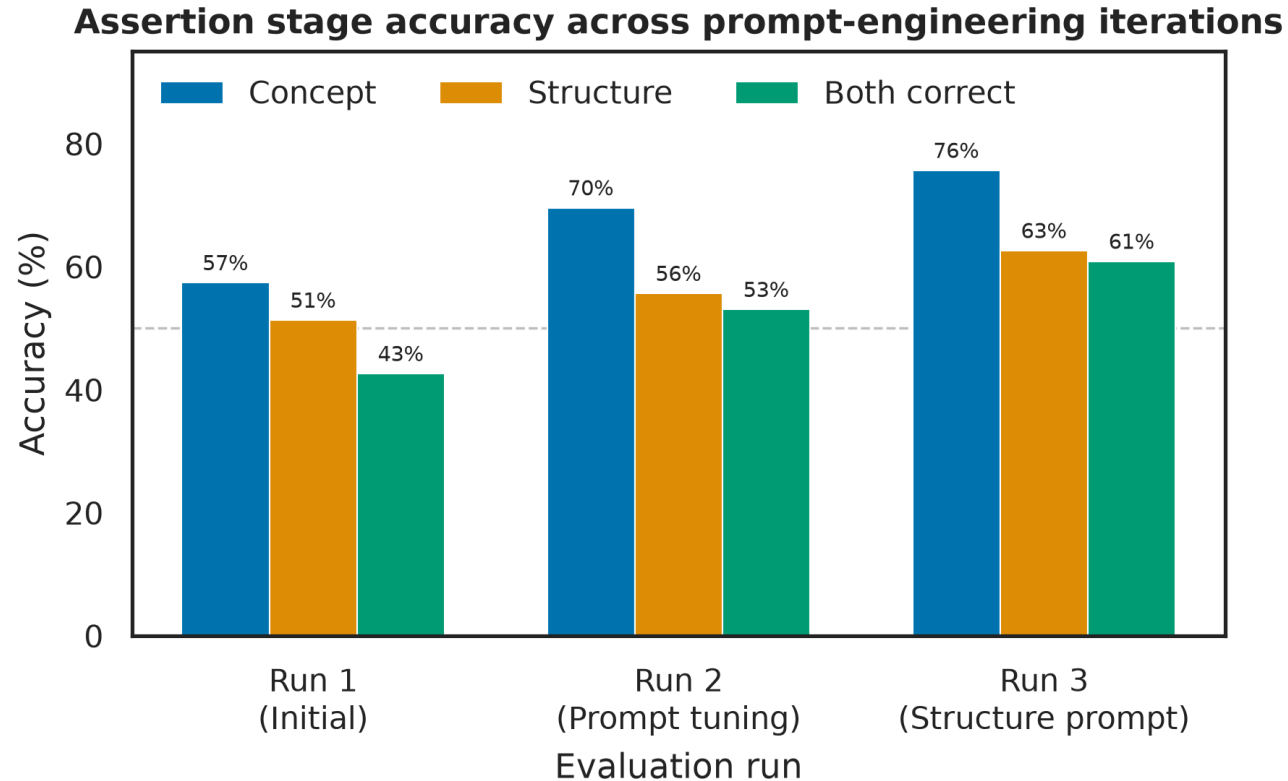
Note:

Exact string match understates quality. "How satisfied are you with this product?" vs "...with your product?" are both perfect, but don't match exactly.

→ **Run 4** adds an **LLM-as-judge** alignment score (1–5) that measures meaning, not strings.

KEY FINDINGS

Three iterations of prompting: steady gains



Concept

57% → 70% → **76%**

Structure

51% → 56% → **63%**

Both correct

43% → 53% → **61%**

+18pp concept, +11pp structure, with **no training**, just better prompts & scoring.

KEY FINDINGS: WHAT DROVE THE GAINS

What each iteration changed

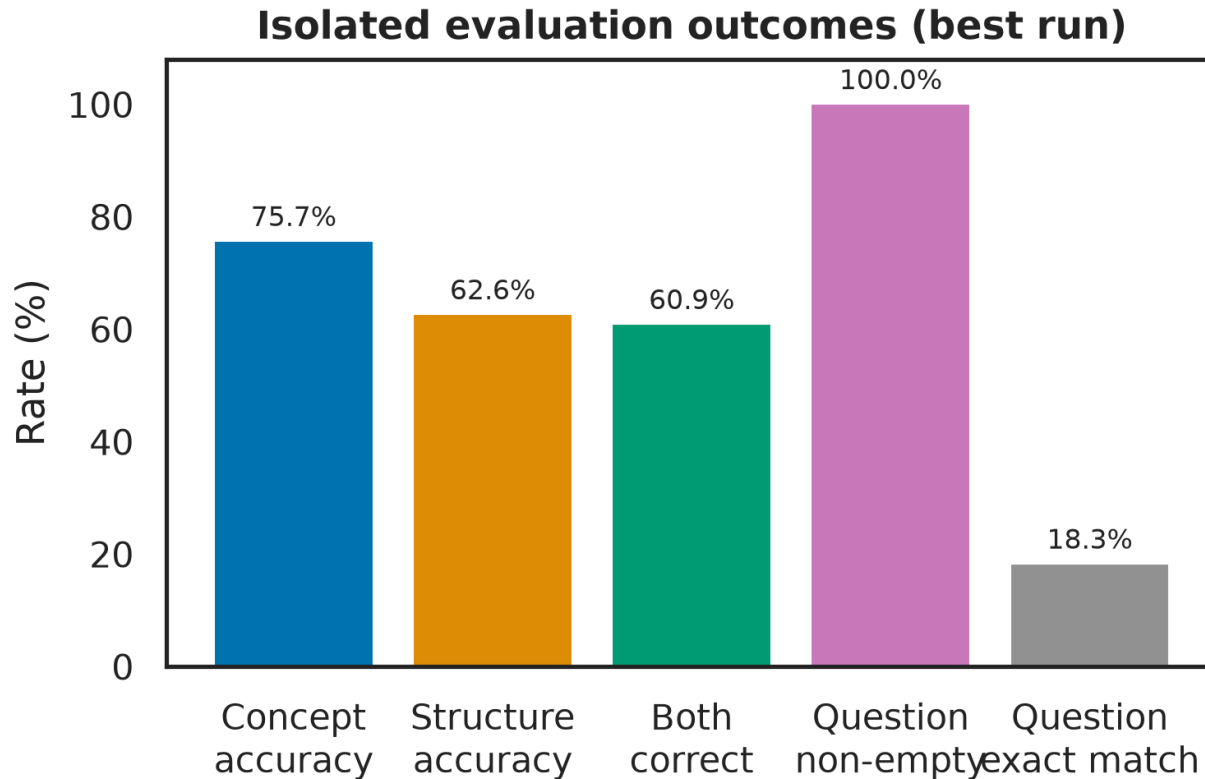
Run	What we changed	Concept	Structure
1 · Initial	Plain zero-shot prompt with the rulebook	57%	51%
2 · Prompt tuning	UK/US spelling normalization · concept <code>enum</code> constraint · disambiguation table + worked examples for confused concepts	70%	56%
3 · Structure focus	Structure-specific examples (<code>xFD</code> / <code>xPRy</code> ...) · JSON repair for malformed output · question-format fix	76%	63%
4 · + Judge	Same outputs as Run 3 · adds LLM-as-judge semantic scores (config flag only, no prompt/model change)	= 76%	= 63%

Each change targeted the previous run's biggest error bucket

Run 1's spelling + confusion errors → fixed in Run 2. Run 2's structure-code gap → targeted in Run 3. **Error-driven, not random tweaking.**

KEY FINDINGS: BEST RUN

Where we landed



75.7%
CONCEPT ACCURACY

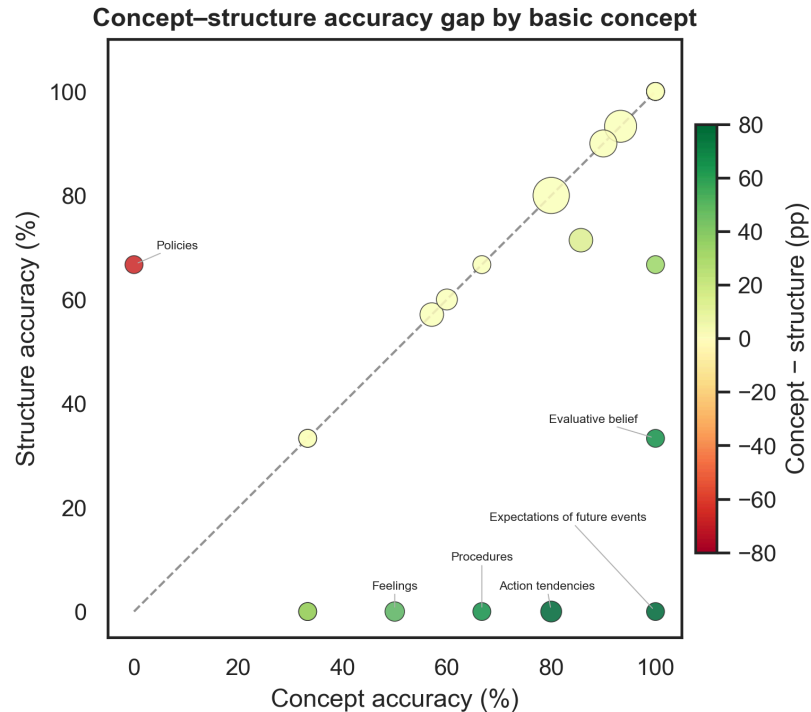
62.6%
STRUCTURE ACCURACY

100%
QUESTION COVERAGE

Question generation **always** produces a sensible question. Exact-match is only 18%, but that's the **metric's** strictness, not bad questions (see examples).

KEY FINDINGS: THE CORE INSIGHT

Structure codes are the bottleneck

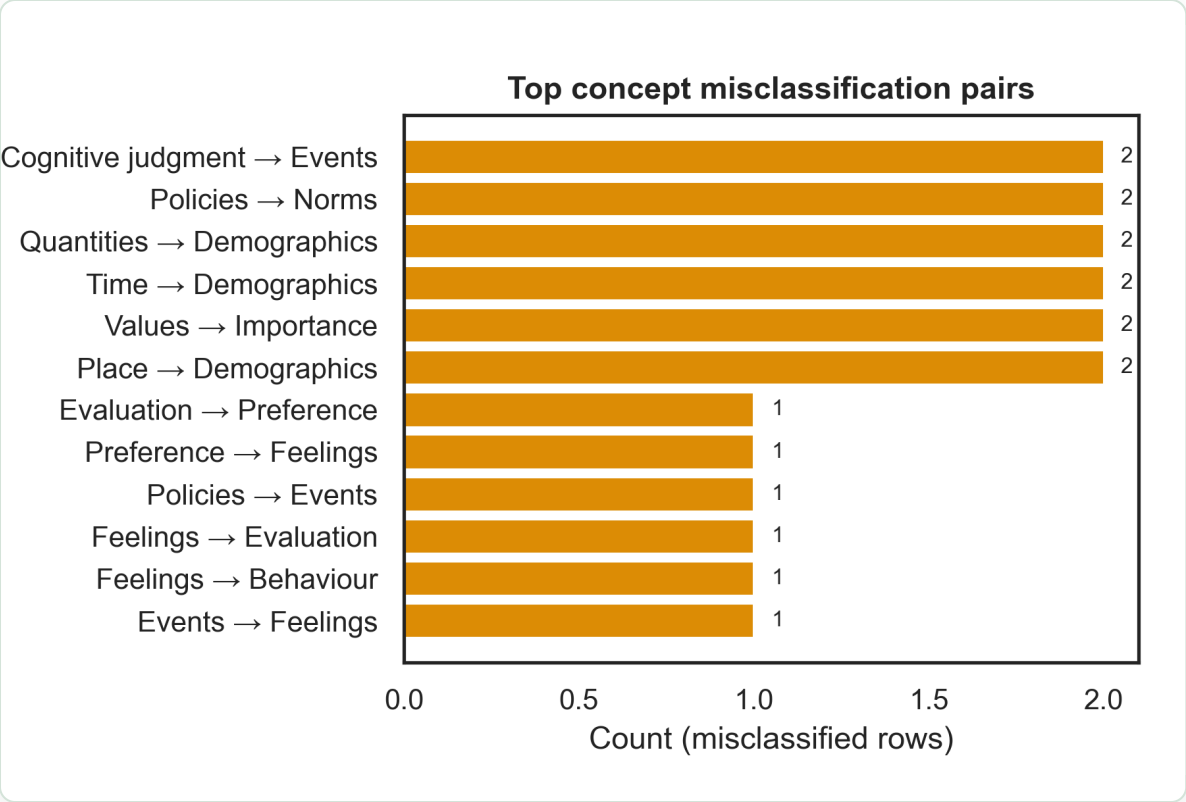


Takeaway: Identifying **what** is measured is largely learnable by prompting; pinning the exact **grammar code** is the remaining hard problem.

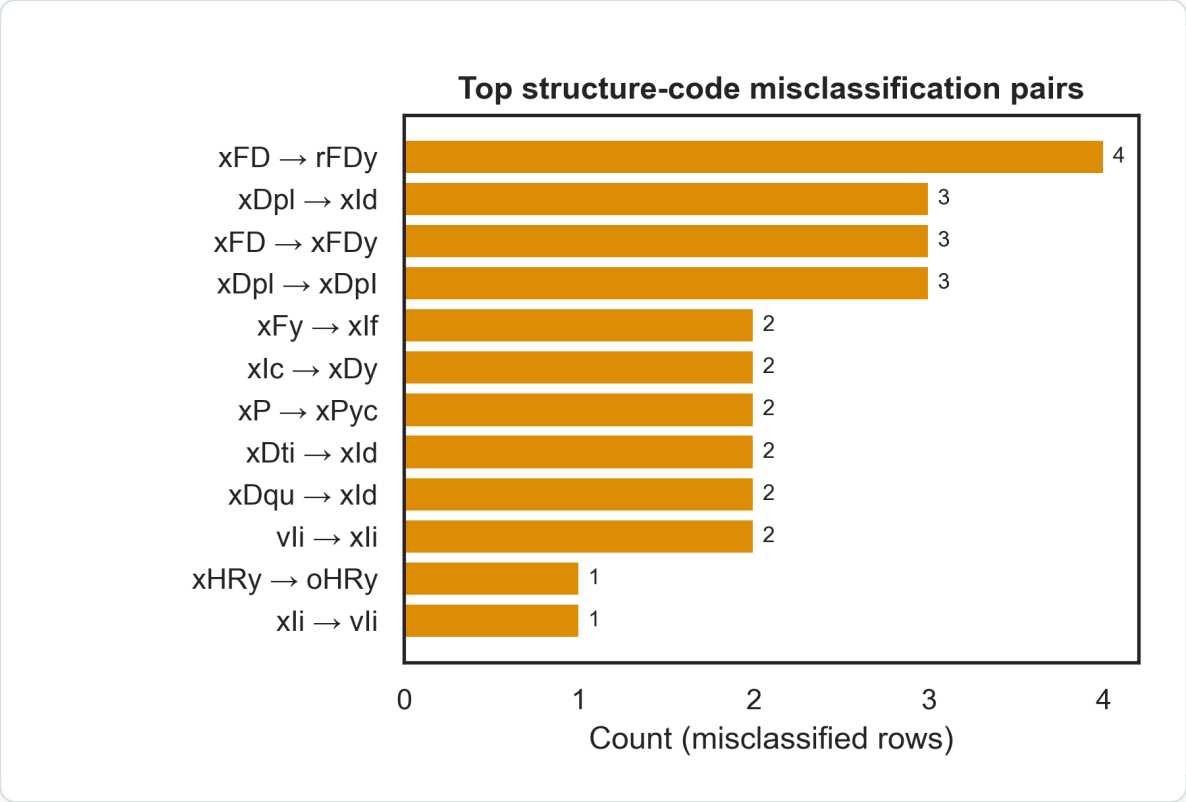
- Points **below the diagonal** = concept right, but structure wrong.
- Whole concepts get the category right yet score **0% on structure: Expectations of future events, Action tendencies, Feelings, Procedures.**
- Structure codes are **fine-grained and theory-laden**: e.g. intention (`rFDy`) vs future event (`xFD`), distinctions even experts debate.

KEY FINDINGS: ERROR ANALYSIS

Where it goes wrong (and why it's reasonable)



Concept confusions



Structure-code confusions

Cognitive judgment→Events, Place/Quantities/Time→Demographics, Policies→Norms · structure: **xFD→rFDy** (future event vs intention), **xPRy↔xIpr**. These are **semantically adjacent**: the model lands one cell away, not wildly off.

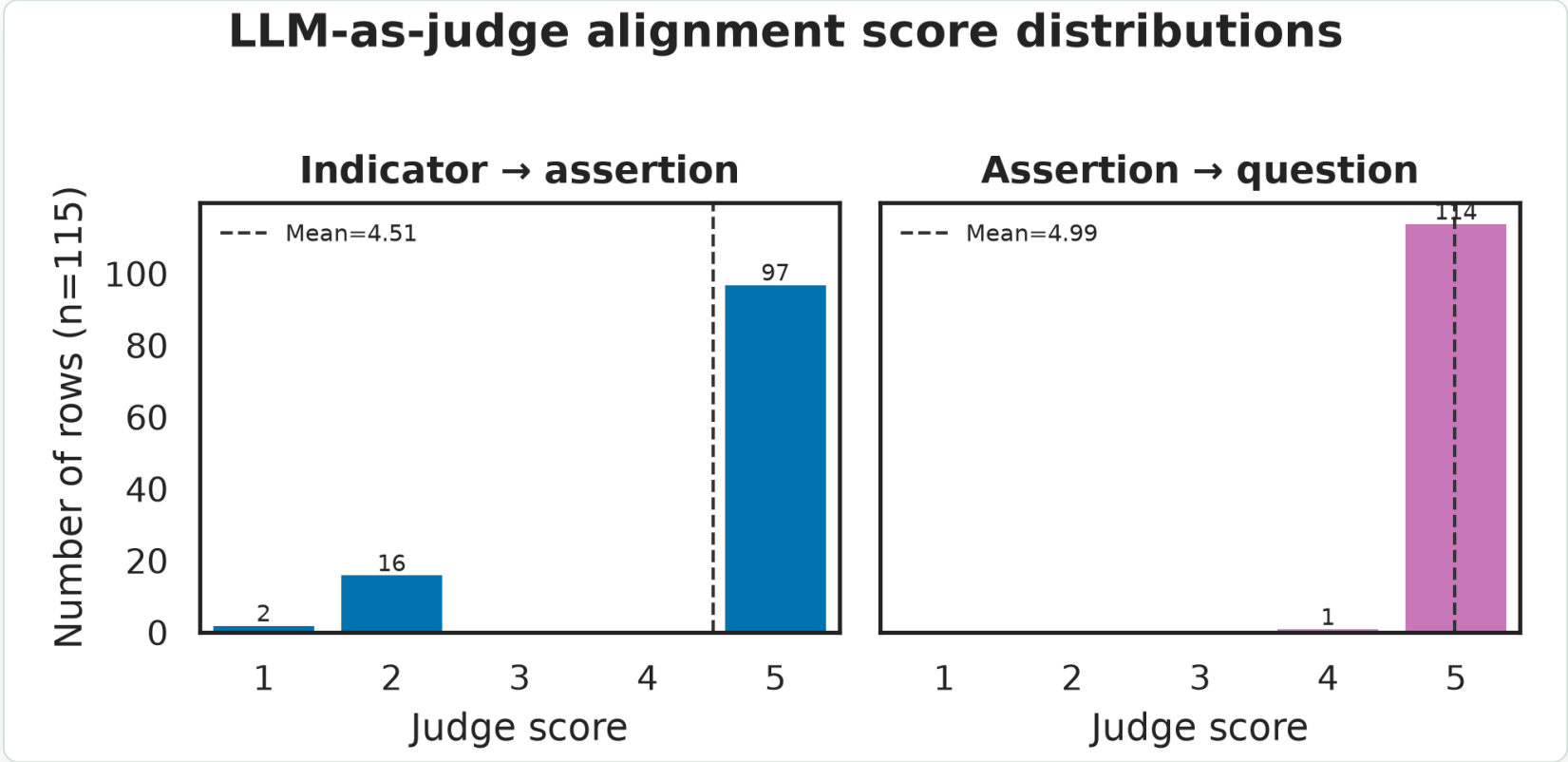
KEY FINDINGS: CONCRETE EXAMPLES

What the outputs actually look like

Indicator	Model output	Verdict
Overall satisfaction with product	Evaluation · xIe · "I am satisfied with this product." → "How satisfied are you with this product?"	concept ✓ structure ✓
Repurchase intention	Action tendencies · rFDy (gold xFD) → "How likely are you to purchase... again?"	concept ✓ structure x
Immediate safety concerns	Feelings (gold Events), latched onto the word "concerns"	concept x

Assertions and questions are **fluent and plausible**; remaining errors are **taxonomy nuances**, not broken language.

LLM-as-judge: semantic alignment scores (1–5)



84%

INDICATOR → ASSERTION = 5

99%

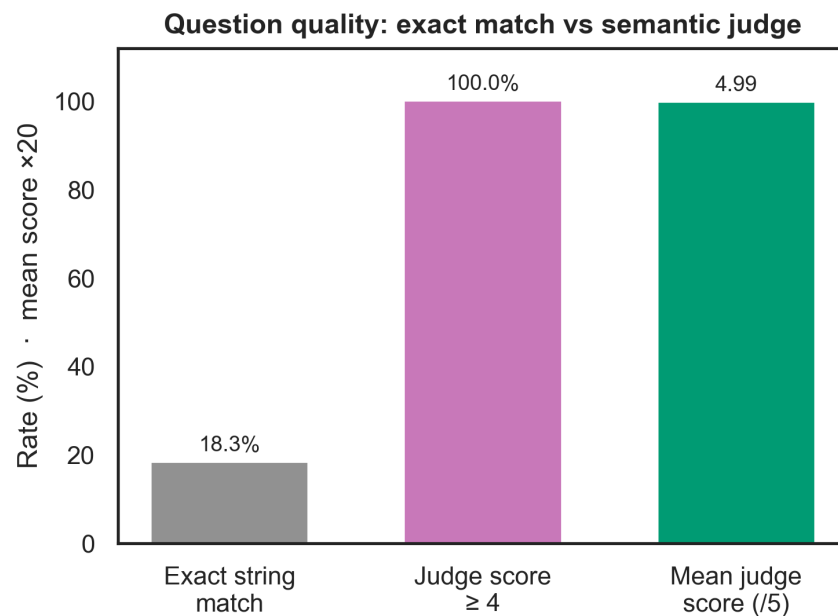
ASSERTION → QUESTION = 5

99%

ROWS SCORED ≥ 4

A judge model rates how faithfully each output captures its input. Same outputs as Run 3: **this adds meaning-based scoring** on top of the exact-match numbers.

Exact match was measuring the wrong thing



Takeaway

The low exact-match was a **metric artefact**, not bad questions. We report **judge alignment** for the question stage.

- Exact match: **18%**. Judge ≥ 4 : **99%**.
- **94 / 94** questions that did **not** match the gold string still scored ≥ 4 for meaning.
- Correlation between exact match and judge: **$r \approx 0.04$** , essentially none.

DISCUSSION

What it means

- An **off-the-shelf 9B model**, with careful prompting and **zero training**, reaches **~76% concept / ~63% structure** on a 22-way expert taxonomy: a strong baseline.
- **Concept identification** is largely solvable by prompting + good examples + output constraints.
- **Structure assignment** is the genuine frontier: fine distinctions, rare codes, theory-dependent.
- **Question generation** is essentially solved: the judge scores it **4.99/5**; the 18% exact match was a metric artefact, not a quality problem.
- Even when the **taxonomy label** is wrong, assertions still read as faithful paraphrases (judge 4.21 vs 4.61 when correct); taxonomy errors $\neq$ broken meaning.
- **One catch**: the judge is the **same model** (self-grading) $\rightarrow$ scores may be optimistic; an external 72B judge is the honest next check.

DISCUSSION

Limitations

Data

115 rows; many concepts at **n = 3** → noisy per-concept estimates. One annotator's scheme.

Metric

Exact-match understates question quality; judge scores exist but are **self-graded**, with no human or external-judge check yet.

Model & runs

Single model, greedy decoding (temp 0), single run → no variance estimates.

Task ambiguity

Some concept/structure distinctions are **inherently debatable**, even for experts: a ceiling on achievable accuracy.

Takeaways

- A prompt-driven pipeline **turns raw indicators into theory-grounded survey items** end-to-end.
- Iterative, error-driven prompting added **+18pp concept accuracy** with no training.
- **Structure codes are the frontier**; the LLM-as-judge confirms questions are excellent (**4.99/5**) despite low exact match.

Next steps

- External judge (Qwen2.5-72B) to remove self-grade bias
- Structure-focused few-shot
- LoRA fine-tuning on the assertion stage
- Expand & balance the gold set

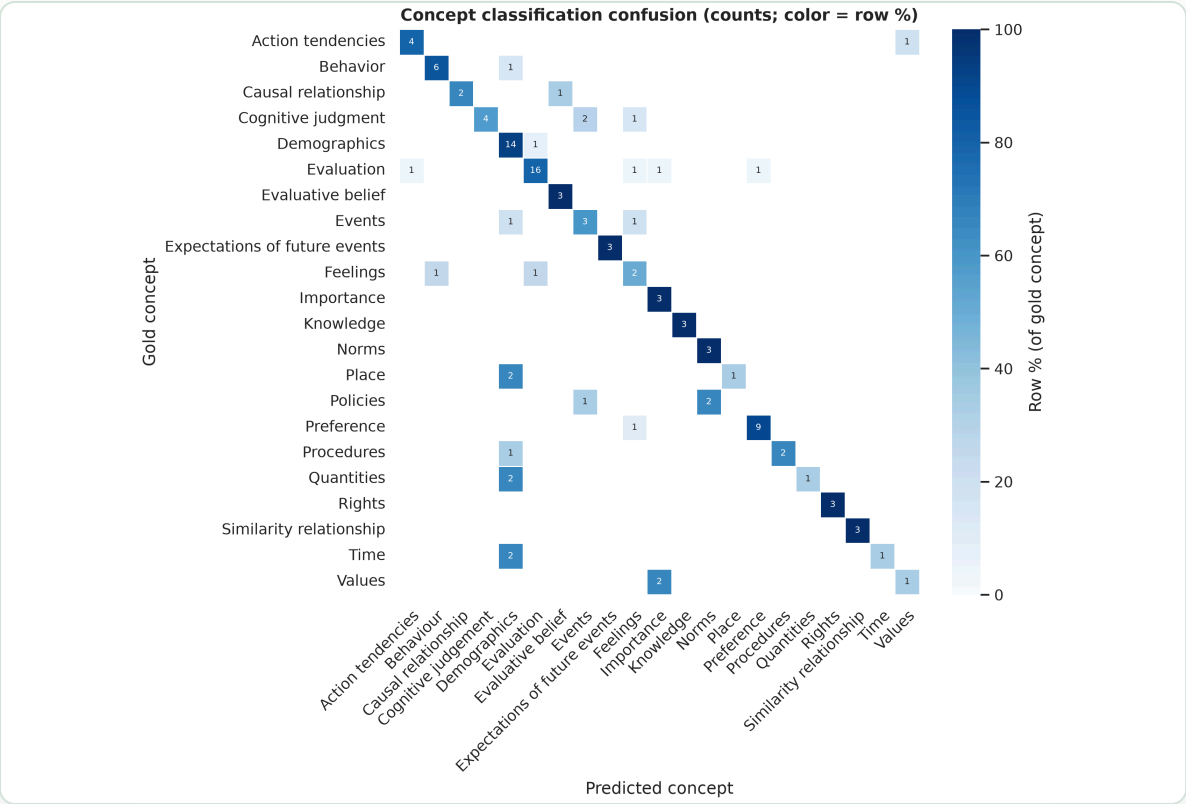
Thank you!

Questions & Discussion.

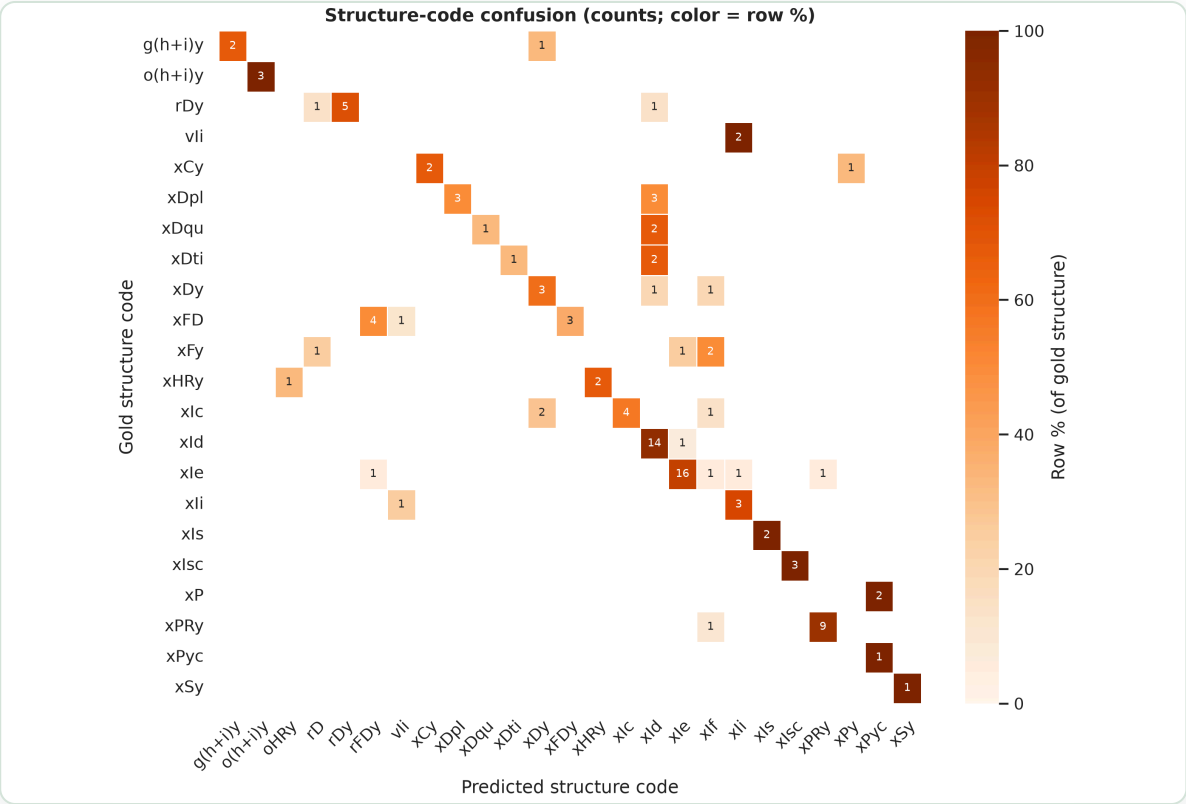
Code: `nlp-css-seminar`

APPENDIX: BACKUP

Full confusion heatmaps



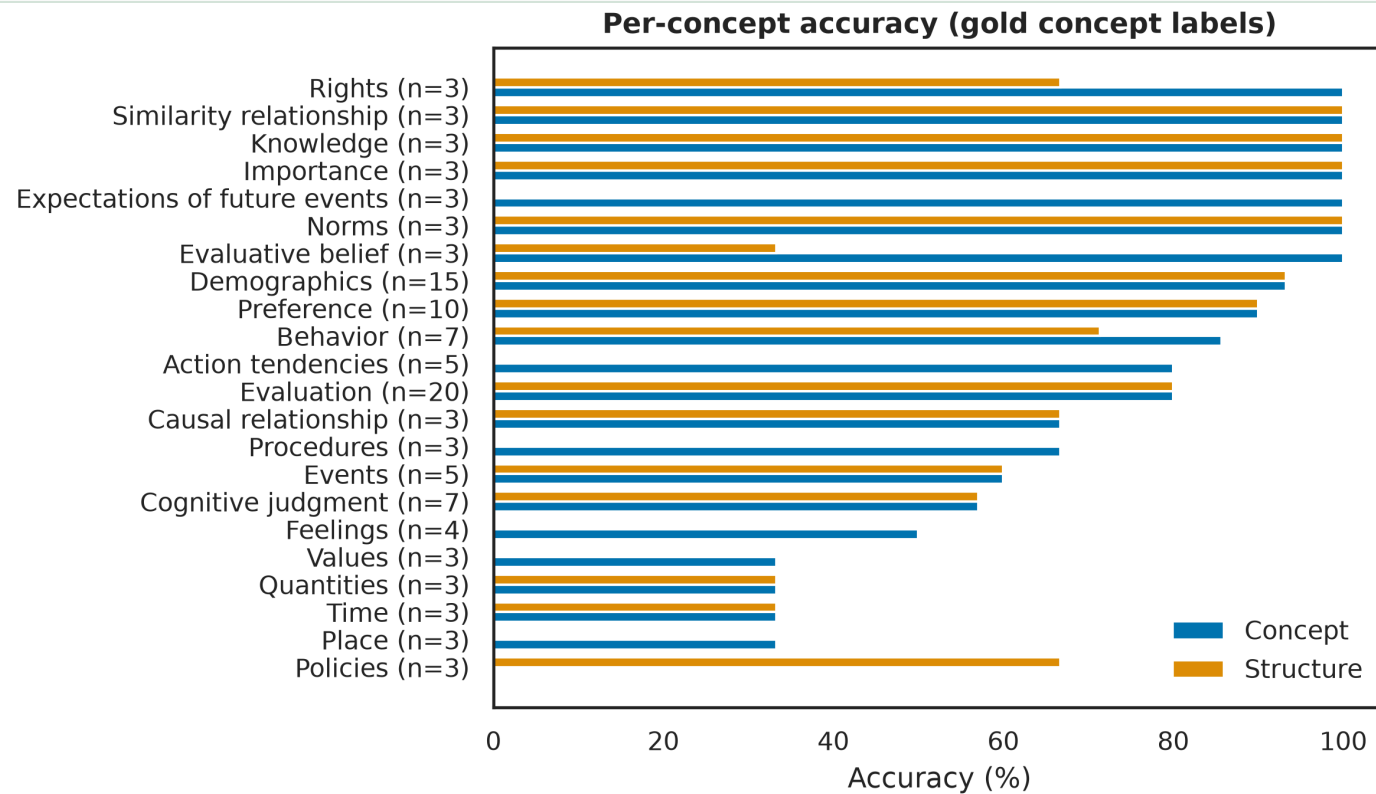
Basic concepts



Semantic structure codes

KEY FINDINGS: BY CATEGORY

Some concepts are easy, some are hard



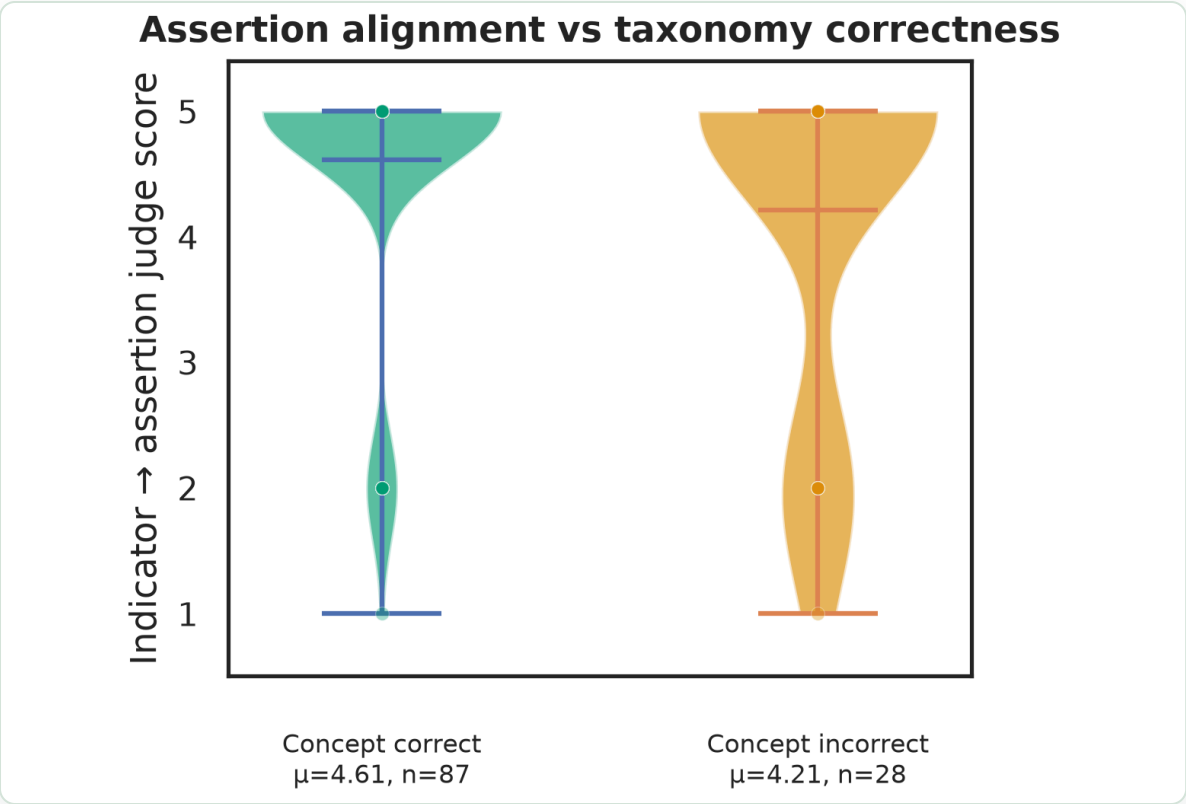
Near-perfect

Evaluation, Demographics, Preference, Importance, Knowledge, Norms, Rights

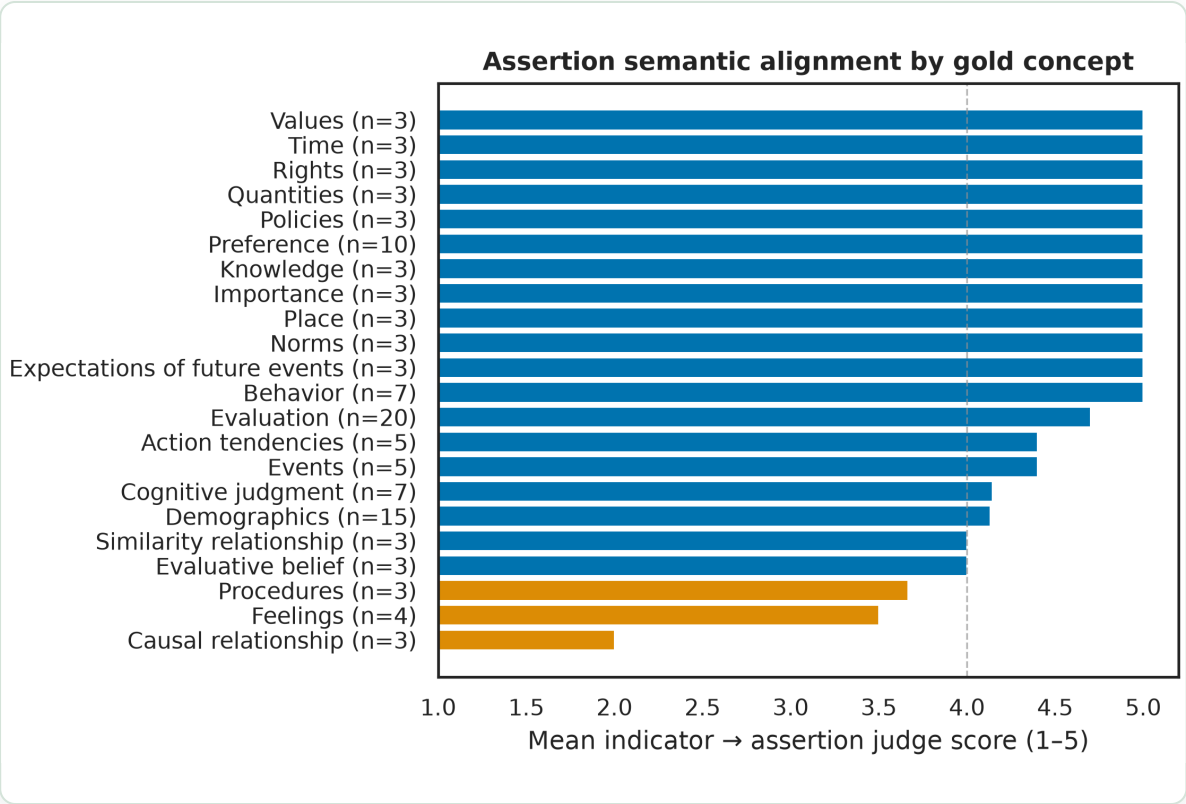
The pattern

Frequent, well-defined concepts are easy; **rare, abstract ones** (Policies, Place, Values, Feelings) are hard.

Judge scores in detail



Assertion alignment when concept is correct vs incorrect



Mean indicator → assertion score by gold concept

APPENDIX: GOLD DATASET SAMPLES

Examples of ground-truth annotations (n=115)

Indicator	Concept	Structure	Assertion	Question
Overall satisfaction	Evaluation	xIe	I am satisfied with this product.	How satisfied are you with your product?
Expectations met	Evaluative belief	xPyc	This product met my expectations.	How well did this product meet your expectations?
Desired improvement	Preference	xPRy	I would prefer a change in a particular aspect of my experience.	What one thing would most improve your experience?
Repurchase intention	Action tendencies	rFDy	I intend to purchase from this organization again.	Likelihood of purchasing from you again?
Emergency contact	Demographics	xId	My emergency contact is [person].	Who is your emergency contact?
Exercise frequency	Behaviour	rDy	I engage in physical exercise regularly.	How often do you engage in physical exercise?

Subjective items focus on internal states/evaluations and typically map to rating scales.

Objective items focus on facts/behaviors and map to frequency or open-ended formats.

ANTICIPATED QUESTIONS

Quick answers to the obvious "why?"s

Question	Answer
Why not GPT-4 / Claude?	Privacy, cost, reproducibility, and it must run on LRZ. An open model is the research-friendly choice, and we keep a stronger model in reserve as the judge .
Why 9B, not 72B?	Fits one GPU → fast iteration. The 72B is earmarked for LLM-as-judge to avoid self-grading bias.
Why isolated evaluation?	So one stage's errors don't contaminate the other, letting us localise the bottleneck (the assertion stage).
Is 76% good?	For a 22-way expert taxonomy with zero training , yes, and it was still climbing each iteration.
Isn't the judge the model grading itself?	Yes, and scores may be optimistic because of that. We flag it and plan an external 72B judge . But judge vs exact-match correlate at $r \approx 0.04$, confirming exact match misses genuine quality.